



BY KIRAN MAZUMDAR-SHAW

A Tech Reboot to Ensure Healthcare for All

To serve the needs of the 1.64 billion people expected to live in India by 2047, the country needs to leverage its expertise in tech



MY VISION FOR India in 2047 envisages a healthcare landscape that is both advanced and affordable. Not just for a few, but for the masses. Not just for India, but for the globe.

I foresee a future where a focus on preventive healthcare will proactively prevent illness and bring down the disease burden. Where predictive healthcare will substitute the need for reactionary interventions and where per-

sonalised solutions will form the crux of all therapeutic endeavours.

The convergence of health tech, digital, media and mobile telecommunications will completely revolutionise India's healthcare system by the time the country celebrates the 100th year of its Independence.

Convergence of IT and life sciences is making an impact: The convergence of information technology and life sciences is creating a more resilient, agile and innovative healthcare system in India and across the globe. We already see signs of this happening. The eSanjeevani telemedicine service had served 76 million patients by December 1, 2022. The Indian SARS-CoV-2

Genomics Consortium, or INSACOG, had sequenced over 250,000 SARS-CoV-2 genomes till November 2022. Recent advances in AI applications have enabled clinicians to rapidly interpret radiology exams like X-rays and CT scans. Over 80 per cent of healthcare professionals in the country are using electronic health records (EHRs) in their practice leading to better quality of care and patient outcomes. Portable diagnostics available on smartphones and tablets are enabling our frontline health workers and technicians in early detection of diseases such as breast and oral cancers. These examples illustrate that we are at the cusp of an interesting revolution where the future of medicine and biotech belongs.

Technological excellence helped India tackle Covid-19: The tectonic shifts caused by Covid-19 brought the Indian biotechnology and biopharma industry into the global limelight. Thanks to the intensive use of technology, India had administered over 2 billion doses of Covid-19 vaccines till November 2022. Technology led to exponential growth in our testing capacity from one test conducted in the lab at the National Institute of Virology (NIV), Pune, in January 2020 to nearly a billion (0.9 billion) Covid-19 tests conducted up to November 30, 2022 (per ICMR data).

Reimagining health in India in the next 25 years: The healthcare of the future will stand on three pillars: Preventive, Predictive, and Personalised. In 2047, healthcare can look something like this for a person living in a remote village in Ladakh. A person having a persistent cough for a week will be able to launch an app on their smartphones and speak into it. This app, using speech recognition and NLP (natural language processing) algorithms, would then ask the person to cough and enquire about symptoms. In the background,